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Emotional Heatmap 2

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Emotional Heatmap

This Help Center article describes the Emotional Heatmap utilizing eye tracking and Facial Expression Analysis.

The Emotional Heatmap processes eye gaze and facial expressions and visualizes them as a heatmap across a group of respondents. The visualization can be utilized to tell what elements on your tested content earn the most visual attention at the same time tell the facial expression response to that element.

Prerequisites:

Supported stimulus types: Images and Surveys.

Images include image stimulus and gaze-mapped reference images generated from iMotions Gazemapping.

Supported sensors: It requires both eye tracking and facial expression analysis data.

For facial expression analysis data, it will only work for AFFDEX. If only one or none of these two signals is available, the Emotional Heatmap will not be generated.

Eye tracking data from both iMotions Desktop study and the RDC (remote data collection) module will work.

Supported iMotions version: iMotions 9.3.21 and later

What is an Emotional Heatmap?

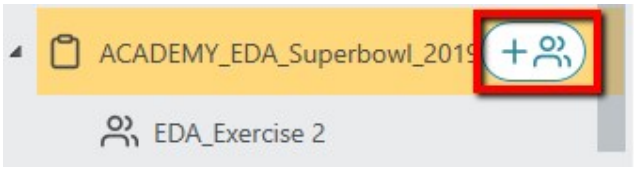
The iMotions Emotional Heatmap is generated in a similar way to the eye tracking Heatmap. The gaze data from the eye tracking Heatmap is filtered with a threshold based on the Facial Expression Metrics that the user has selected. The filtered gaze points are then visualized and color coded, based on the selected colors. Metrics, threshold values, and colors are customizable by the user and will be explained in detail in the following section. The generated Emotional Heatmap will allow researchers to pinpoint emotional activity in response to the visual elements that were gazed at in the tested content.

How to generate and export an Emotional Heatmap in iMotions?

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If you have eye tracking and facial expression analysis (AFFDEX) data available in your study, then you are ready to go. First, follow the steps below to create an Analysis in iMotions:

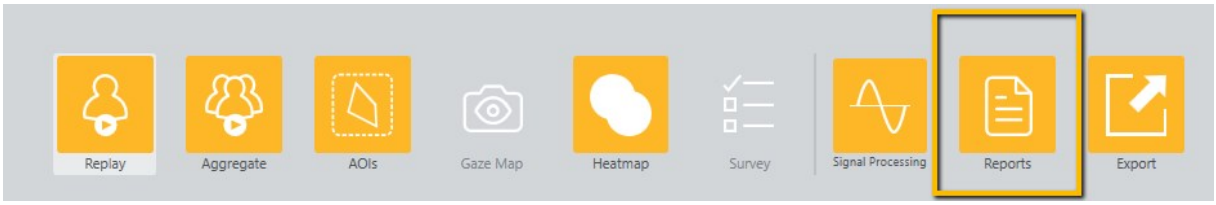
1. Create a new analysis for the study for which you wish to run the Emotional Heatmap:



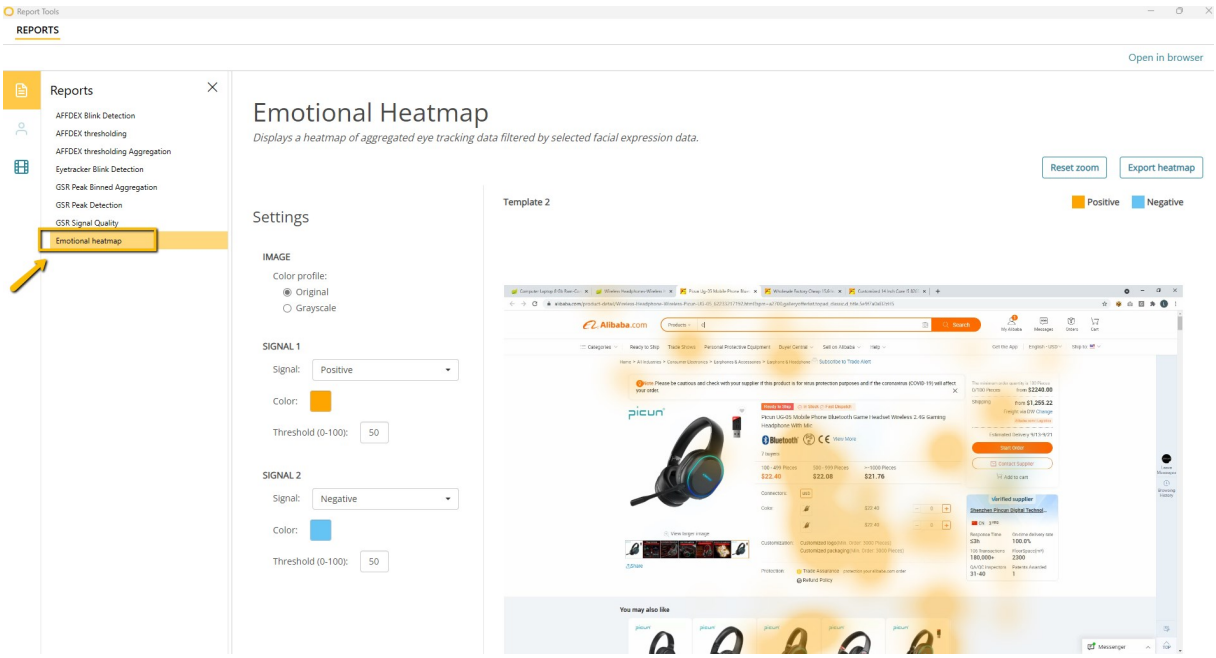
2. Advance through the steps "Segmentation settings" and "Segmentation criteria" with your preferred selections and continue until you reach the step "Signal processing". The respondents you included in this step will be used to generate Emotional Heatmap later. Please note that for the Emotional Heatmap, you do not need to add any algorithms under "Active flows". You can click "Next" to proceed.

3. Select the stimuli to be included in the analysis. Then click the "Add" button to add the analysis and close the dialog.

4. The Emotional Heatmap can then be generated by clicking the "Reports" button:



Navigate to the "Emotional heatmap" tab listed together with the other Reports:



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Please note that in the left side panel, you won't be able to switch between respondents since the analysis is run at *aggregation* level. But you are able to switch between stimuli.

Now you can click and choose to customize your own Emotional Heatmap. The options are listed below:

- **IMAGE, Color profile:** The "Original" and "Grayscale" allows you to keep the original colors in the stimulus, or use a gray scale instead.
- **SIGNAL 1, SIGNAL 2:** The dropdown list contains all the metrics from AFFDEX. One can choose up to two metrics of interest and their colors to be shown on Emotional Heatmap. The "Threshold" will decide if a detected gaze point is visible or not on the Emotional Heatmap, based on its paired AFFDEX metrics data: If the AFFDEX metrics data falls below the "Threshold", then the corresponding gaze point will not be included in the heatmap; the other way around, if the AFFDEX metrics data is above the "Threshold", then the corresponding gaze point will be included.
- **Zoom:** It's possible to left-click and drag with the mouse on the Emotional Heatmap to zoom in. Click "Reset zoom" to zoom out and restore to its original size.
- **Export heatmap:** Click "Export heatmap" to export the generated Emotional heatmap in .png format outside iMotions.

Questions and Troubleshooting Tips:

***Q:** Why can't I find the "Emotional heatmap" tab after I click "Reports", or see error messages on my Emotional Heatmap iMotions page (see screenshot below)?*

A: Please double-check if your study collected both eye tracking data AND AFFDEX data. Only when having both of these signals, you will be able to generate Emotional heatmap properly.

Example when the study only collected AFFDEX data but without eye tracking data:

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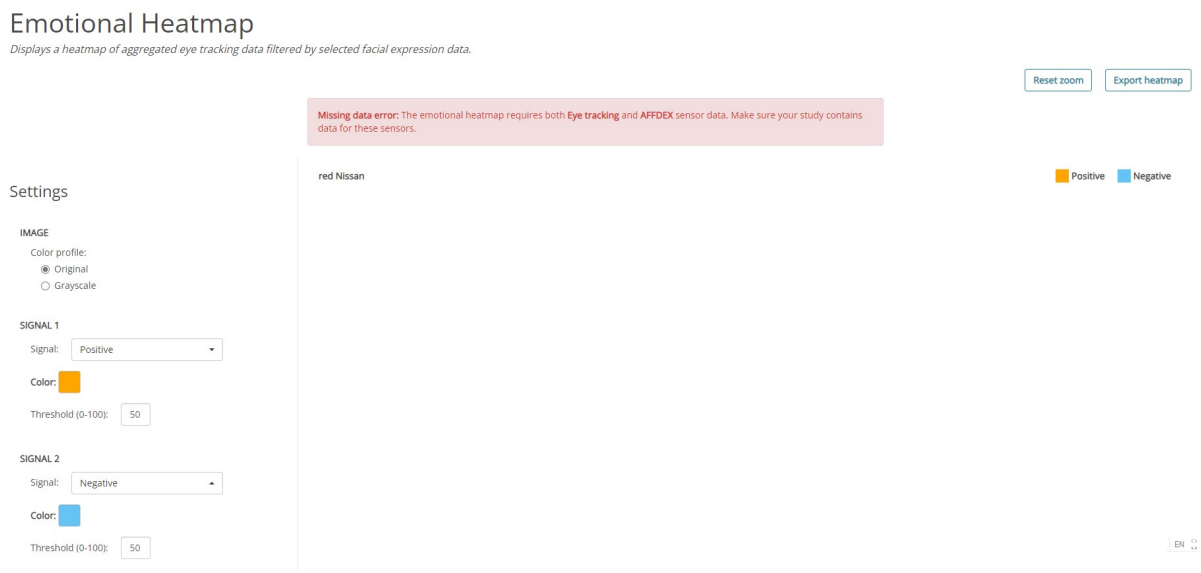


Image Caption

Q: *Why do I only see three metrics (Positive, Negative, Neutral) after I click the Signal drop-down?*

A: Please wait until the loading is complete, then you will be able to see the full list of the metrics from AFFDEX.

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